Program 4: Working with Selenium

1.Sauce.demo webpage

Step 1: create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**

```
shreya@ShreyaM:~$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -Dint
eractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shreya
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shreya/MyMavenSeleniumApp01
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  32.878 s
[INFO] Finished at: 2026-03-17T22:37:01+05:30
[INFO] ----------------------------------------------------------------------------
```

Step 2: Before we connect to web servers we need to have chrome driver installed. To download chrome driver

To check the chrome version type the command **google-chrome –version** in the terminal

```
shreya@shreyam:~$ google-chrome -version
Google Chrome 145.0.7632.116
```

extract the version and store it in a variable CHROME_VERSION

Check if ChromeDriver is installed correctly:

```
shreya@shreyam:~$ chromedriver --version
ChromeDriver 145.0.7632.117 (de4e5d6e13b995f5e04137970c0e8ec456017467-refs/branch-heads/7632@{#3101})
```

Step 3 : edit the pom.xml file and add the dependencies

The pom.xml has the code

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp01</artifactId>
  <packaging>jar</packagi<ng>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp01</name>
  <url>http://maven.apache.org</url>
```

```
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.29.0</version>
</dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
          <groupId>org.apache.maven.plugins</groupId>

          <artifactId>maven-jar-plugin</artifactId>
          <version>3.0.1</version>
         <configuration>
             <archive>
                <manifestEntries>
                   <Main-Class>com.example.App</Main-Class>
                </manifestEntries>
             </archive>
          </configuration>
```

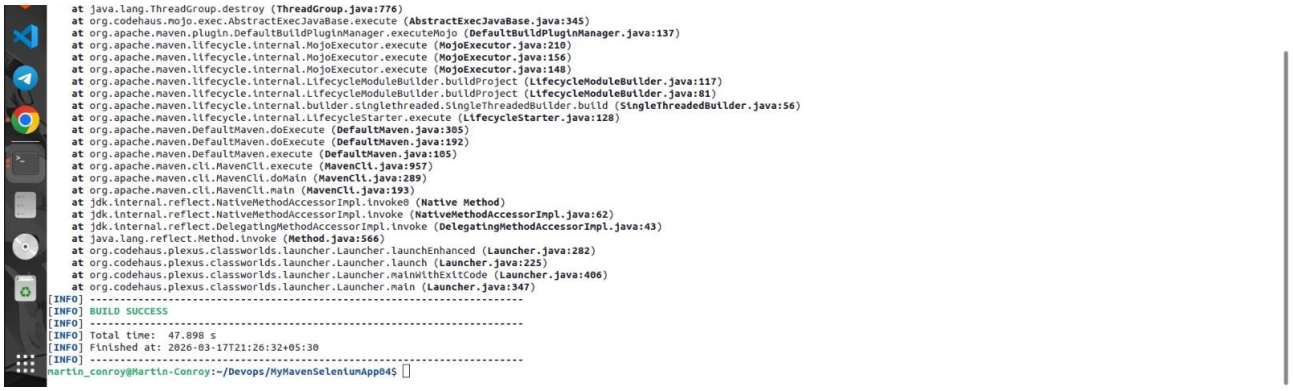
```
        </plugin>

</plugins>
</build>
</project>
```

Step 4: Edit the App.java

```
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
public class App
{
    public static void main(String[] args) throws InterruptedException
    {
        WebDriver driver = new ChromeDriver();
        Thread.sleep(2000);
        driver.get("https://www.saucedemo.com/");
        Thread.sleep(2000);
        driver.manage().window().maximize();
        Thread.sleep(2000);
        driver.findElement(By.id("user-name")).sendKeys("standard_user");
        Thread.sleep(2000);
        driver.findElement(By.id("password")).sendKeys("secret_sauce");
        Thread.sleep(2000);
        driver.findElement(By.id("login-button")).click();
        Thread.sleep(2000);
    }
}
```

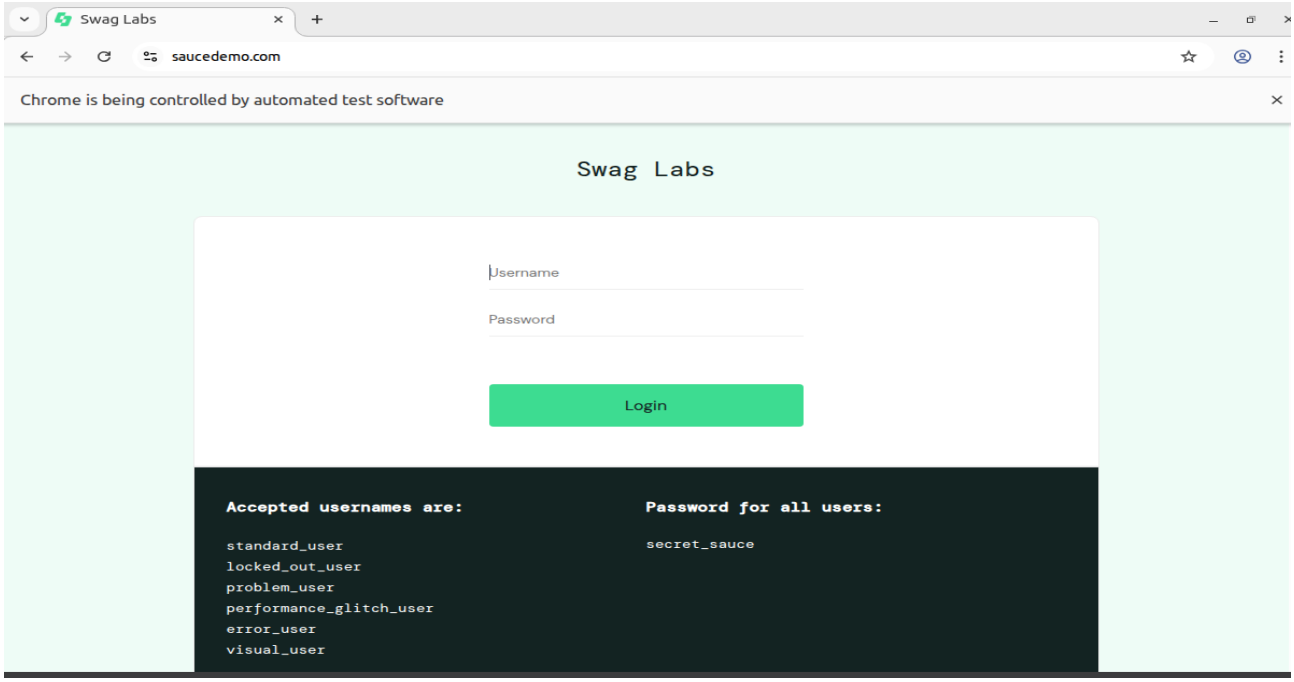
Step 5:build and run the maven application

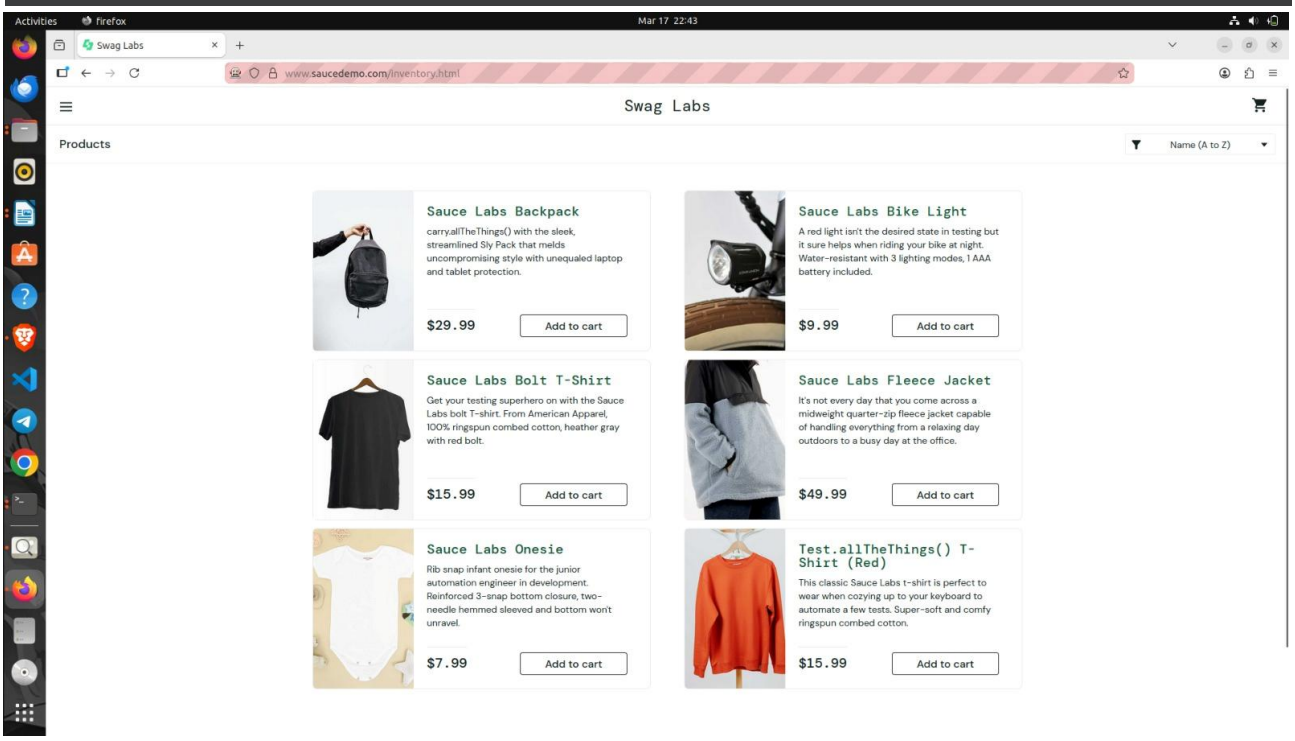
Type **mvn clean install** in the terminal



To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"**
it opens saucedemo.com and we can login using the credentials

Devops


Swag Labs
saucedemo.com
Chrome is being controlled by automated test software
Swag Labs
Username
Password
Login
Accepted usernames are:
standard_user
locked_out_user
problem_user
performance_glitch_user
error_user
visual_user
Password for all users:
secret_sauce


Activities
firefox
Mar 17 22:43
Swag Labs
www.saucedemo.com/inventory.html
Swag Labs
Products
Name (A to Z)
Sauce Labs Backpack
carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.
$29.99
Add to cart
Sauce Labs Bike Light
A red light isn't the desired state in testing but it sure helps when riding your bike at night. Water-resistant with 3 lighting modes, 1 AAA battery included.
$9.99
Add to cart
Sauce Labs Bolt T-Shirt
Get your testing superhero on with the Sauce Labs bolt T-shirt. From American Apparel, 100% ringspun combed cotton, heather gray with red bolt.
$15.99
Add to cart
Sauce Labs Fleece Jacket
It's not every day that you come across a midweight quarter-zip fleece jacket capable of handling everything from a relaxing day outdoors to a busy day at the office.
$49.99
Add to cart
Sauce Labs Onesie
Rib snap infant onesie for the junior automation engineer in development. Reinforced 3-snap bottom closure, two-needle hemmed sleeved and bottom won't unravel.
$7.99
Add to cart
Test.allTheThings() T-Shirt (Red)
This classic Sauce Labs t-shirt is perfect to wear when cozying up to your keyboard to automate a few tests. Super-soft and comfy ringspun combed cotton.
$15.99
Add to cart

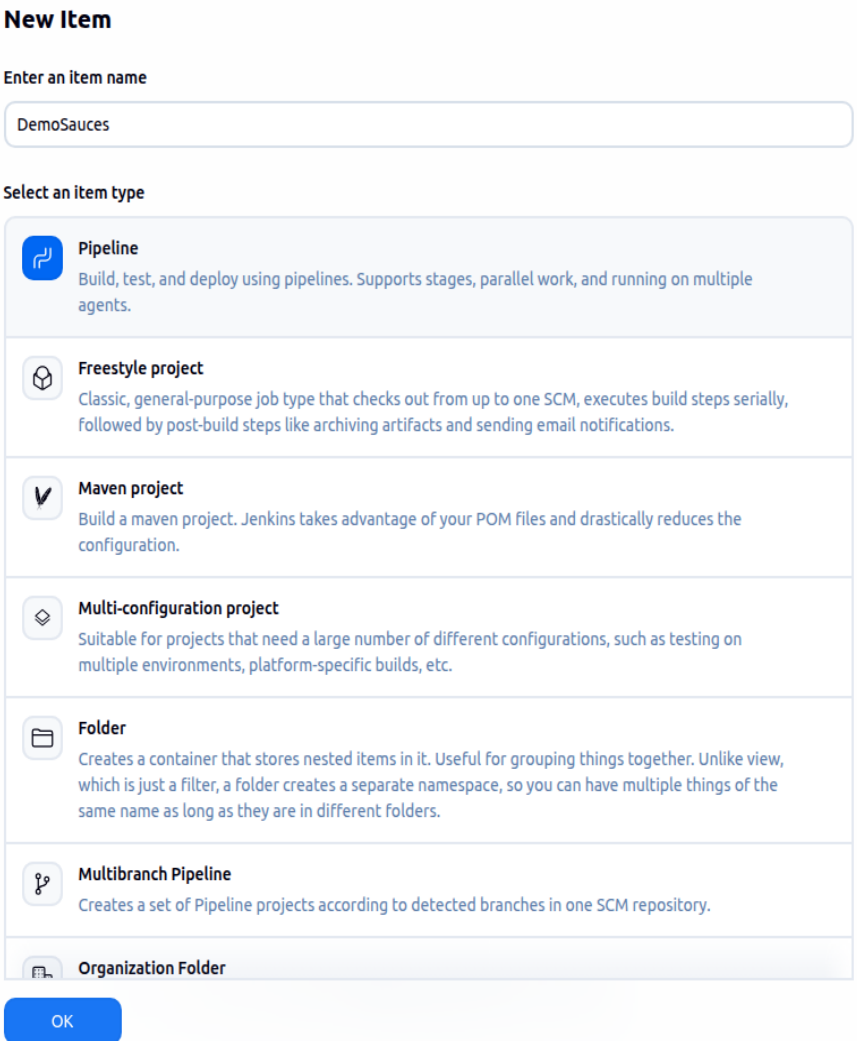
SauceDemo in jenkins:

Step 1: in the terminal type the code saucedemo to enter the folder .
Type the command gedit JenkinsFile to open the Jenkinsfile and paste the code

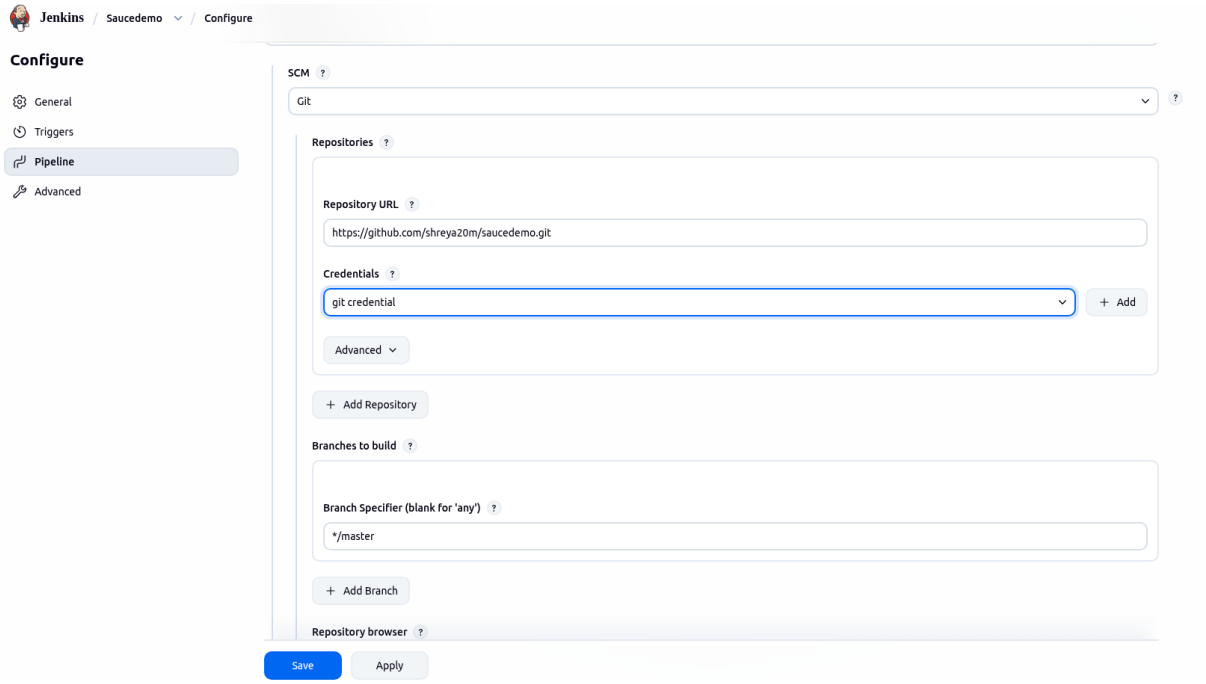
```
pipeline {
    agent any

    tools {
        maven 'Maven'
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/shreya20m/saucedemo.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package -DskipTests'
            }
        }

        stage('Run Application') {
            steps {
                sh 'java -jar target/saucedemo-1.0-SNAPSHOT.jar'
            }
        }
    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
```

```
            echo 'Build failed!'
        }
    }
}
```

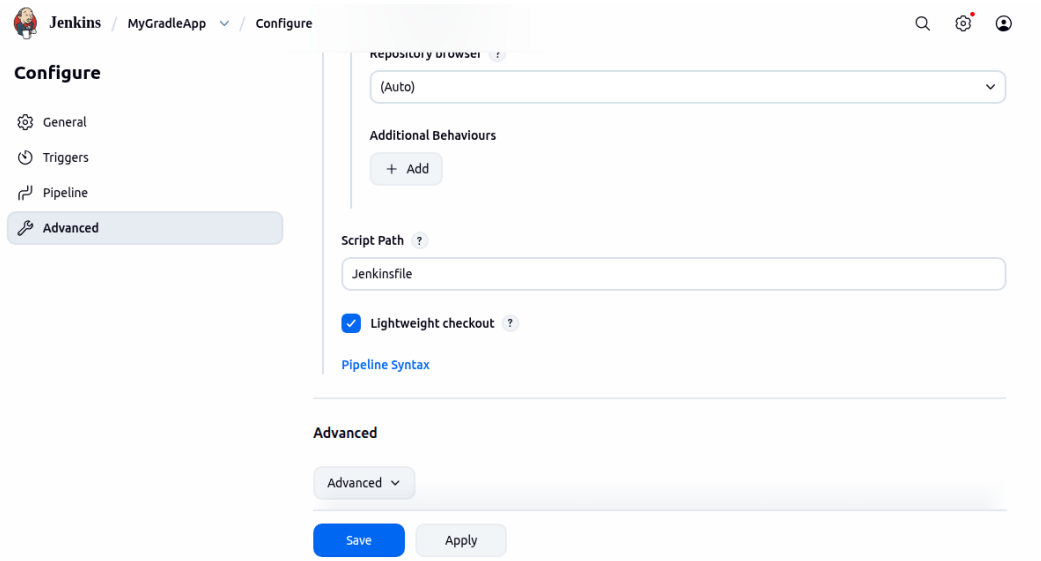
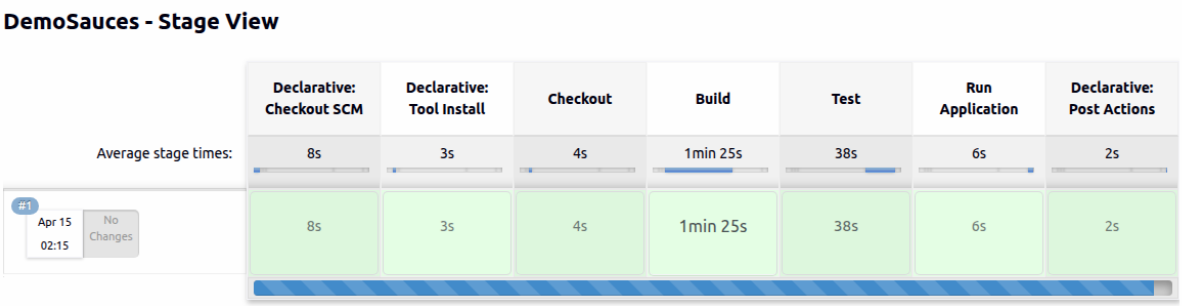
Step 2: in Jenkins Dasboard click on New Item
Give the name Saucedemo and select Pipeline



In configure scroll down to Pipeline and choose pipeline script from SCM,give the repository url and choose GitHub Credential



Then click on apply and save



Step 3: go to Jenkins Dasboard select the Saucedemo pipeline and select Build Now. After build select Full Stage View. Once the errors are corrected Build again